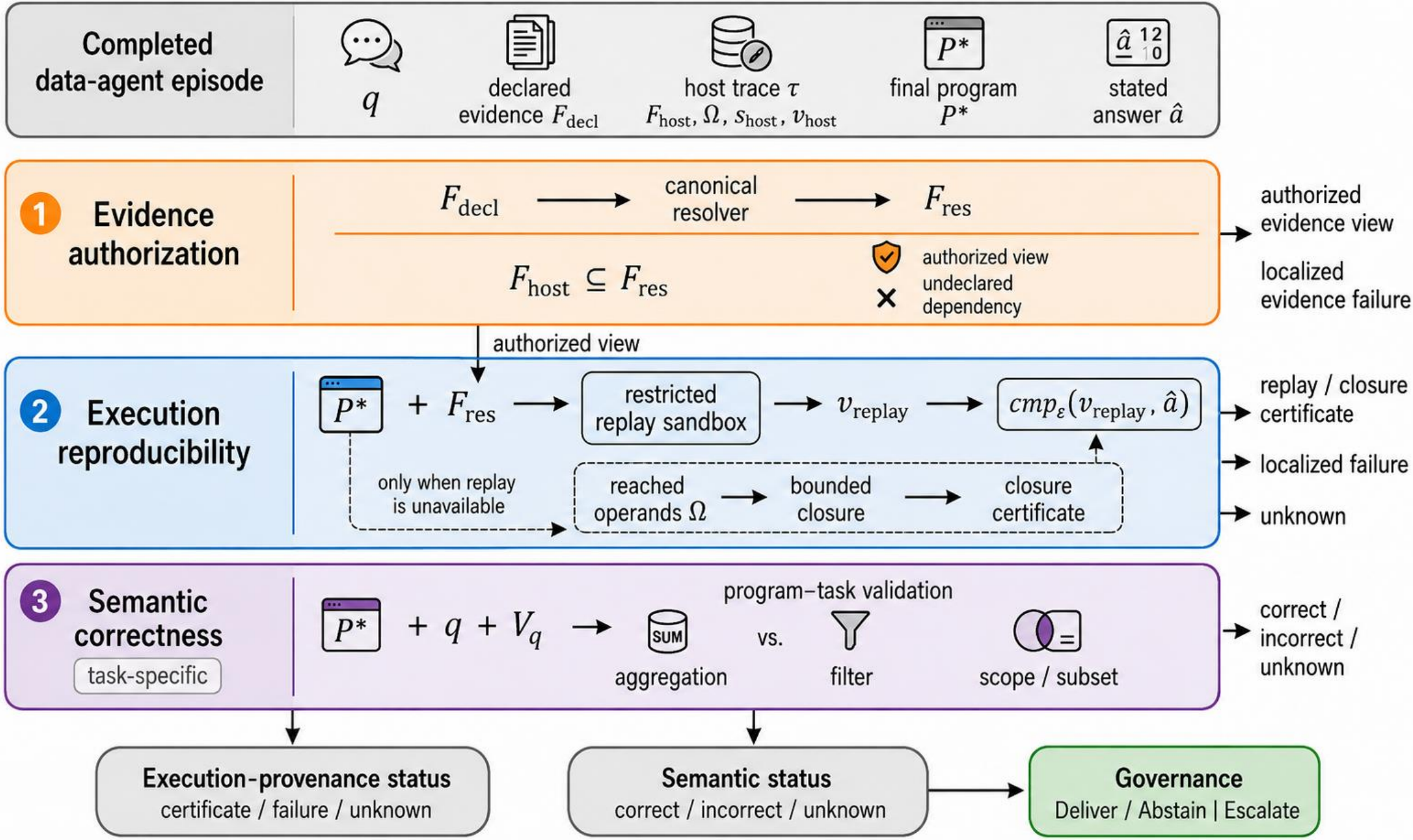

Completed data-agent episode
q
declared evidence F_{decl}
host trace τ $F_{\text{host}}, \Omega, s_{\text{host}}, v_{\text{host}}$
final program P^*
stated answer $\hat{a}$
1 Evidence authorization
F_{decl}
canonical resolver
F_{res}
$F_{\text{host}} \subseteq F_{\text{res}}$
authorized view
undeclared dependency
authorized evidence view
localized evidence failure
authorized view
2 Execution reproducibility
P^* + F_{res}
restricted replay sandbox
v_{replay}
$cmp_{\varepsilon}(v_{\text{replay}}, \hat{a})$
only when replay is unavailable
reached operands Ω
bounded closure
closure certificate
replay / closure certificate
localized failure
unknown
3 Semantic correctness
task-specific
P^* + q + V_q
program–task validation
SUM
aggregation
vs.
filter
scope / subset
correct / incorrect / unknown
Execution-provenance status
certificate / failure / unknown
Semantic status
correct / incorrect / unknown
Governance
Deliver / Abstain | Escalate